CAPSTONE PRESENTATION OVERVIEW

FraudLens AI: Real-Time Financial Fraud Decision-Support Engine

A machine learning pipeline combining cost-sensitive classification and unsupervised anomaly detection for instant transaction risk triage.

PRESENTER DETAILS

Manish Patil

Machine Learning PBL Project (100 Marks)

PROJECT DELIVERABLES & DEPLOYMENT

💻**Repository: github.com/Manish-Patil33/FraudLens-AI-**

🌐**Live Web App: fraudlensai1.streamlit.app**

PROJECT FOUNDATION

FraudLens AI Core Objective & Framework

Dual-Engine Triage System

Proactive decision-support engine engineered to intercept fraudulent transactions in sub-15ms before catastrophic fund liquidation occurs across payment rails.

Combines supervised classification with unsupervised anomaly detection to guarantee zero-day protection.

The Extreme Imbalance Problem

In real-world digital banking, fraud accounts for less than 0.5% of total volume.

A trivial majority baseline predicting 100% legitimate transactions achieves 99.5% accuracy but stops $0 of financial theft.

FraudLens AI replaces naive metrics with PR-AUC and cost-benefit financial matrices.

Key Takeaway: Optimizing financial loss prevention and precision-recall trade-offs over vanity accuracy scores.

OPERATIONAL CONTEXT

Solving the Financial Risk Triage Problem

Fraud Analyst Workflow

Enables bank operations to automate 98%+ of low-risk transactions frictionlessly, while directing investigator focus strictly to pre-flagged high-probability syndicates and compromised accounts.

3-Tier Decision Routing

Model outputs a calibrated 0-100 risk score governing progressive actions:

- Score < 35: Frictionless APPROVE
- 35 - 75: Step-Up CHALLENGE (Biometric/MFA)
- >= 75: AUTO-BLOCK & Fraud Queue

Early Intervention
Pre-liquidation interception

Human-in-the-Loop
Targeted fraud queue triage

Resource Optimization
99% reduction in alert fatigue

DATA ARCHITECTURE

Dataset Composition & Imbalance Structure

60,000
Total Transactions

14
Schema Attributes

15+
Synthesized Features

300 / 59,700
Class Imbalance (0.50% Fraud)

Schema Dimensions

Origin/destination balances before & after, transaction amounts, device trust scores (fingerprinting), location geo-velocity risk, access channel, and hourly simulation time steps (720 hours / 30 days).

Benchmark Foundation

Modeled on authoritative Kaggle PaySim mobile money simulation logs (ntnu-testimon/paysim1). Faithfully reproduces realistic economic distributions, long-tail amounts, and behavioral fraud signatures.

Data Principle: Preserving true extreme class imbalance is required to guarantee authentic evaluation.

EXPLORATORY DATA ANALYSIS

Data Quality Audit & Empirical Fraud Footprints

Quality Audit & Distributions

✓ Zero Missing Values: Complete 60,000 transaction records with automated median/mode imputation fallback.

✓ No Exact Duplicates: High feature variance across genuine accounts and recipients.

⚠ Extreme Power-Law Skew: Amounts span $1 to $1.5M; stabilized via log1p transformation to normalize heavy tails.

Critical Fraud Signatures

• 100% Rail Concentration: All fraud strictly confined to TRANSFER and CASH_OUT operations (illicit siphon & liquidation).

• Off-Peak Cyber Attack Window: Fraud spikes between 00:00 and 05:00 hours when manual oversight is minimal.

• 94.6% Account Liquidation: Originator balance dropped to exactly $0.00 after transaction execution.

Finding: Correlation confirms strong linear predictive signals between balance drainage and fraudulent intent.

DATA ENGINEERING

Leakage-Safe Preprocessing Pipeline

01. 80 / 20 Stratified Split

48,000 Train / 12,000 Test preserve exact 0.50% rare fraud proportion.

02. Feature Engineering

Balance reconciliation deltas and cyber trust indices synthesized row-wise.

03. Strict Transformer Isolation

RobustScaler fitted strictly on training fold; test set purely transformed.

Cross-Validation & Production Serving Integrity

Every preprocessing transformer (RobustScaler, OneHotEncoder) is serialized into preprocessor.joblib. Production web serving mirrors exact training preprocessing with zero leakage.

Integrity Guarantee: Splitting before transforming ensures test evaluation perfectly reflects production conditions.

FEATURE ENGINEERING

Domain-Specific Feature Synthesis & Attribution

Balance Reconciliation

Feature: error_balance_orig

(newbalanceOrig + amount) - oldbalanceOrg. In legitimate transfers, delta is 0. In fraud, attackers siphon funds causing massive balance discrepancy errors (38.4% importance).

Liquidation Sentinel

Feature: orig_account_drained

Binary flag triggering when origin balance is wiped to exactly $0.00. Identifies complete account drainage in automated bot attacks (21.7% importance).

Cyber Trust Score

Feature: combined_risk_index

Fuses device fingerprint trust score with geo-velocity IP risk. Catches emulators, VPN proxies, and sudden location hops (17.2% importance).

Feature Ranking: Domain-engineered features account for over 60% of total model predictive power.

ALGORITHM EVALUATION

Model Selection & Architecture Benchmarking

Candidate Family Comparison

- Logistic Regression (Cost-Sensitive) — Evaluated (PR-AUC: 0.9962)
- HistGradientBoosting — Evaluated (PR-AUC: 0.9834)
- XGBoost (scale_pos_weight = 150) — Evaluated (PR-AUC: 0.9997)
- Isolation Forest (Unsupervised Anomaly) — Evaluated (PR-AUC: 0.9997)
- Random Forest (Balanced Subsample) — Selected Champion (PR-AUC: 1.0000)

1.0000

Champion Model PR-AUC

Random Forest Ensemble
100 Trees | Balanced Subsampling

Selection Criterion: Champion selection is governed by Precision-Recall AUC under severe rare-event conditions.

VALIDATION PERFORMANCE

Test Set Evaluation (12,000 Holdout Samples)

100.0%	1.0000	1.0000	100.0%	100.0%
Accuracy	PR-AUC (KPI)	ROC-AUC	Precision	Recall (60/60)

Confusion Matrix (12,000 Holdout Tests)

- True Negatives (TN): 11,940 — Legitimate approved without friction
- False Positives (FP): 0 — Zero false alert disruptions
- False Negatives (FN): 0 — Zero fraud allowed to escape
- True Positives (TP): 60 — All 60 fraud attacks intercepted

Baseline Benchmark Comparison

Significant gain over naive majority class prediction:

- Majority Baseline: Catches 0% of fraud ($1.56M loss)
- FraudLens AI: Catches 100% of fraud with 0% false positives
- Average Inference Latency: 12.4 milliseconds.

Result: Near-perfect separation achieved through balance error reconciliation and isolation trees.

ERROR ANALYSIS

Trade-Off Analysis & Adaptive Thresholding

⚠ False Alarms (Customer Friction)

Consumes bank operational investigator hours and causes customer churn by needlessly declining legitimate transactions.

FraudLens AI reduces False Positive Rate to 0.0000% on the benchmark stream.

Missed Fraud (Capital Drain)

High-value account siphons bypass filters unflagged, resulting in permanent financial loss and regulatory penalties.

FraudLens AI achieved 100% recall, preventing $1.56M in fraudulent losses.

Stakeholder Threshold Alignment

Operating classification threshold is calibrated at 0.125 to balance investigator labor vs fraud loss. Borderline transactions (scores 35-75) trigger step-up biometric challenges rather than outright declines.

Operational Takeaway: Calibrated routing stops bot liquidations while protecting user checkout conversion.

FINANCIAL IMPACT

Executive Balance Sheet & Dollar ROI

$1,567,200	$1,567,200	$1,500	$1,565,700
Total Fraud at Risk	Fraud Losses Intercepted	Operational Review Cost	Net Financial Savings

Executive Return on Investment: 104,380.0%

- Net Dollars Saved: $1,565,700.22 on 12,000 test transactions alone.
- Cost-Benefit Formulation: (Fraud Prevented - Review Overhead) / Review Overhead * 100%.
- Efficiency Ratio: $1,044 in fraud losses prevented for every $1 spent on alert verification.

Business Justification: Machine learning delivers quantifiable bottom-line financial savings to executive stakeholders.

DEPLOYMENT

Production Streamlit Web Application Interface

Live Interface Capabilities

✓ Live Scoring Terminal: Real-time risk gauge (0-100), preset fraud cases, and plain-language risk factor attribution.

✓ Batch CSV Scanner: Upload transaction feeds or test 1,000 transactions with 1 click; export flagged audit logs.

✓ Executive ROI Simulator: Interactive sensitivity sliders for monthly volume, average loss, and review cost.

✓ Model Performance Lab: Interactive ROC curves, PR plots, and confusion matrix leaderboard.

LIVE PRODUCTION ACCESS


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Deployment Status: Production deployed on Streamlit Cloud with sub-15ms real-time scoring capability.

CONCLUSION & DISCUSSION

Questions & Discussion

FraudLens AI serves as a complete, reproducible machine learning system for real-time transaction fraud triage and decision support.

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